aman gupta X-B roll 31

1) a 2) b 3) a 4) b

5) a 6) b 7) b 8) c

Q10. $\tan 45 = 1$, $\cos 30 = \sqrt{3}/2$, $\sin 60 = \sqrt{3}/2$

$2 \times 1 + 3/4 - 3/4$

$= 2$

Q11. $A - B = 30°$ ($\sin 30 = 1/2$)

$A + B = 60°$ ($\cos 60 = 1/2$)

$2A = 90$ $A = 45°$

$B = 15°$

Q.13

$(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2$

$= \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta + \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta$

$= 2\sin^2\theta + 2\cos^2\theta = 2(1) = 2$

Q14. $\cos\theta = 12/13$

$\sin\theta = \sqrt{(1 - 144/169)} = \sqrt{(25/169)} = 5/13$

$\tan\theta$ ~~$= 5/12$~~ $12/5$

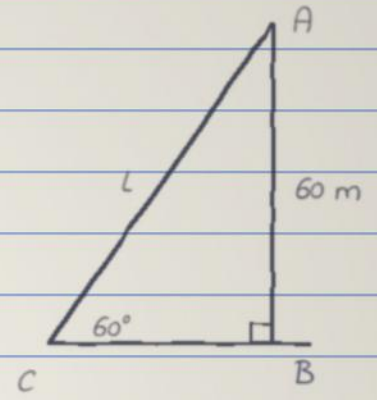
$\cot\theta = 5/12$

$\text{cosec}\theta = 13/5$

$$\sin 60 = \frac{60}{l}$$

$$\frac{\sqrt{3}}{2} = \frac{60}{l}$$

$$l = \frac{120}{\sqrt{3}} = 40\sqrt{3}\ \text{m}$$



Q18.

$\tan 45 = 7/x$

$1 = 7/x \quad x = 7\text{m}$

$\tan 60 = 7/CE$

$\sqrt{3} = 7/CE \quad CE = 7/\sqrt{3} = 4.04$

$\text{height} = 7 + 4.04 = 11.04\ \text{m}$

